



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

EE2352-1 – BATTERY STORAGE AND FUEL CELLS FOR ELECTRIC VEHICLES
(For EE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The specific energy of a storage device is defined as:
A) Energy per unit cost
B) Energy per unit mass
C) Energy per unit charge
D) Energy per unit volume
2. Which of the following is the main advantage of ultracapacitors in HEVs?
A) High power density and long cycle life
B) High specific energy
C) Low cost and light weight
D) High self-discharge rate
3. In hybrid storage, the battery provides
A) Peak power during acceleration
B) Energy for long-duration driving
C) Energy only during regenerative braking
D) Voltage regulation
4. Flywheels store energy in the form of:
A) Electrochemical energy
B) Mechanical rotational energy
C) Electrical field energy
D) Magnetic field energy
5. Which battery chemistry is most sensitive to deep discharge?
A) Lithium-ion
B) Nickel-cadmium
C) Lead-acid
D) Nickel-metal hydride
6. In the equivalent circuit model, the capacitor element is mainly used to represent:
A) Battery's self-discharge
B) Dynamic voltage response
C) Internal resistance
D) State of health
7. Nickel-metal hydride (NiMH) batteries are considered better than lead-acid for HEVs because they:
A) Are cheaper and lighter
B) Have higher specific energy and cycle life
C) Have lower charging efficiency
D) Do not require thermal management
8. In HEV applications, battery packs are usually rated in terms of
A) Ampere-hours and cost per kWh
B) Energy density only
C) Power density only
D) Voltage efficiency only
9. The open-circuit voltage of a hydrogen-oxygen fuel cell at standard conditions is approximately:
A) 0.6 V
B) 0.9 V
C) 1.2 V
D) 1.8 V
10. Which fuel cell type is most suitable for portable and automotive applications due to its low operating temperature?
A) Solid oxide fuel cell (SOFC)
B) Proton exchange membrane fuel cell (PEMFC)
C) Molten carbonate fuel cell (MCFC)
D) Phosphoric acid fuel cell (PAFC)
11. In a fuel cell polarization curve, the activation region corresponds mainly to:
A) Kinetic losses at the electrodes
B) Ohmic resistance of the electrolyte
C) Mass transport limitations
D) Crossover of reactants

12. Non-hydrogen fuel cells can operate on fuels such as:
 - A) Methanol
 - B) Ethanol
 - C) Natural gas
 - D) All of the above
13. In a fuel cell hybrid electric drivetrain, the PPS (Peak Power Source) usually consists of:
 - A) Only the fuel cell stack
 - B) Battery or ultracapacitor
 - C) Transmission system
 - D) Hydrogen storage tank
14. Which parameter is most directly influenced by the design of the gear ratio in a fuel cell hybrid drivetrain?
 - A) Fuel consumption
 - B) Vehicle acceleration and gradeability
 - C) Cooling system efficiency
 - D) Hydrogen purity
15. A key advantage of hybridizing a fuel cell with a battery in an HEV is:
 - A) Lower overall vehicle cost
 - B) Smoother load management and improved transient response
 - C) Reduced weight of the vehicle
 - D) Elimination of fuel storage system
16. In the design example of a fuel cell hybrid drivetrain, the motor size is typically verified using:
 - A) Acceleration performance and gradeability tests
 - B) Cost analysis of the powertrain
 - C) Fuel cell thermal efficiency
 - D) Battery charge-discharge cycle test
17. In a hydraulic energy storage system, energy is stored mainly in:
 - A) Flywheel inertia
 - B) Chemical bonds
 - C) Pressurized fluid in an accumulator
 - D) Magnetic field of an inductor
18. Which of the following fuel cell types is most commonly used in automotive applications?
 - A) Alkaline Fuel Cell (AFC)
 - B) Solid Oxide Fuel Cell (SOFC)
 - C) Proton Exchange Membrane Fuel Cell (PEMFC)
 - D) Phosphoric Acid Fuel Cell (PAFC)
19. In a fuel cell-ultracapacitor hybrid system, the ultracapacitor is mainly responsible for:
 - A) Long-duration energy supply
 - B) Handling transient peak power and regenerative braking energy
 - C) Replacing the fuel cell stack
 - D) Hydrogen compression
20. Which feature makes flywheel energy storage systems particularly attractive for hybrid vehicles?
 - A) High energy density
 - B) Zero maintenance
 - C) High power density and fast response
 - D) Low manufacturing cost

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List the main types of electrochemical batteries used in EV/HEV applications. Write down their typical nominal cell voltage and specific energy range.	5	L*1	1	1
	b) What is the role of the electric double layer in ultracapacitors? Why does water decomposition limit the maximum cell voltage?	5	L2	1	1
	c) Discuss the role of hybridization of energy storage systems (HESS) in EVs/HEVs. Compare passive and active configurations and analyze how hybridization improves battery cycle life and system efficiency.	6	L3	2	1,2
2.	a) Write the charging and discharging reactions of a lead-acid battery.	5	L1	1	1
	b) Explain with a neat sketch the equivalent circuit model of an ultracapacitor.	5	L2	2	1,2
	c) Compare the efficiency and cycle life of batteries, ultracapacitors, and flywheels. Analyze why combining these devices in a hybrid energy storage system can better meet the long-term performance requirements of EVs/HEVs.	6	L3	2	1,2

- | | | | | | |
|-------|--|---|----|---|-----|
| 3. a) | With the help of a neat diagram, explain the general configuration of an Electric Vehicle (EV). List its major components and their functions. | 5 | L2 | 1 | 1 |
| b) | List and explain the limitations of ultra-high-speed flywheels as energy storage devices for EVs. | 5 | L2 | 1 | 1 |
| c) | Explain with a schematic the topology of a Battery Energy Storage System (BESS) in HEVs. Mention the role of power electronics and control system. | 6 | L2 | 2 | 1,2 |

Unit – II

- | | | | | | |
|-------|--|---|----|---|---|
| 4. a) | Explain with a block diagram the control strategy of a Fuel Cell Hybrid Electric Vehicle (FCHEV). Mention the main operating modes. | 5 | L2 | 4 | 1 |
| b) | Explain the consumption of fuel and oxidant in a hydrogen-air fuel cell. How are these related to the cell current? | 5 | L2 | 3 | 1 |
| c) | Explain the different types of fuel cell technologies with their electrolyte materials and operating temperature ranges. | 6 | L2 | 3 | 1 |
| 5. a) | With a neat sketch, explain the system characteristics of a fuel cell. How do auxiliaries like compressors and pumps influence efficiency? | 5 | L2 | 3 | 1 |
| b) | Explain POX reforming method of producing hydrogen. | 5 | L2 | 3 | 1 |
| c) | Explain the method of liquid hydrogen storage for fuel cell vehicles. | 6 | L2 | 4 | 1 |
| 6. a) | List the factors that determine fuel and oxidant consumption in a fuel cell. | 5 | L1 | 3 | 1 |
| b) | Demonstrate how hydrogen can be stored using metal hydrides. Discuss its suitability for fuel cell vehicles. | 5 | L3 | 3 | 1 |
| c) | Define the main components of a Fuel Cell Hybrid Electric Drivetrain. List their functions briefly. | 6 | L1 | 4 | 1 |

Unit – III

- | | | | | | |
|-------|---|---|----|---|-----|
| 7. a) | Explain the differences between various energy Storage technologies. | 5 | L2 | 5 | 1 |
| b) | Explain the electrical equivalent circuit of a Supercapacitor. | 5 | L2 | 5 | 1 |
| c) | Using the hydraulic energy storage model, apply the energy recovery principle during load drop. Analyze the role of accumulator and control valves in maintaining system stability. | 6 | L3 | 5 | 1,2 |
| 8. a) | Explain the influence of SOC estimation on battery charging control? | 5 | L2 | 5 | 1 |
| b) | Draw and explain the equivalent circuit model of an ultracapacitor under charging conditions. | 5 | L2 | 5 | 1 |
| c) | Analyze how flywheel inertia and rotational speed affect energy storage capacity. Discuss trade-offs in design. | 6 | L3 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
